INAUGURAL ARTICLE

Carbon dioxide fixation as a central redox cofactor recycling mechanism in bacteria

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The Calvin-Benson-Bassham cycle (Calvin cycle) catalyzes virtually all primary productivity on Earth and is the major sink for atmospheric CO_2. A less appreciated function of CO_2 fixation is as an electron-accepting process. It is known that anoxygenic phototrophic bacteria require the Calvin cycle to accept electrons when growing with light as their sole energy source and organic substrates as their sole carbon source. However, it was unclear why and to what extent CO_2 fixation is required when the organic substrates are more oxidized than biomass. To address these questions we measured metabolic fluxes in the photosynthetic bacterium *Rhodopseudomonas palustris* grown with ^{13}C-labeled acetate. *R. palustris* metabolized 22% of acetate provided to CO_2 and then fixed 68% of this CO_2 into cell material using the Calvin cycle. This Calvin cycle flux enabled *R. palustris* to reoxidize nearly half of the reduced cofactors generated during conversion of acetate to biomass, revealing that CO_2 fixation plays a major role in cofactor recycling. When H_2 production via nitrogenase was used as an alternative cofactor recycling mechanism, a similar amount of CO_2 was released from acetate, but only 12% of it was reassimilated by the Calvin cycle. These results underscore that N_2 fixation and CO_2 fixation have electron-accepting roles separate from their better-known roles in ammonia production and biomass generation. Some nonphotosynthetic heterotrophic bacteria have Calvin cycle genes, and their potential to use CO_2 fixation to recycle reduced cofactors deserves closer scrutiny.

Calvin cycle | hydrogen gas | metabolic flux analysis | nitrogenase | *Rhodopseudomonas*

At least 170 y ago, it became widely recognized that CO_2 could serve as the sole carbon source for plants and certain bacteria (autotrophic growth). Using experimental results collected by a number of scientists in the late 1700s and early 1800s, the German chemist von Liebig successfully argued that plants obtain carbon from CO_2 gas rather than from carbon in the soil (the prevailing theory at the time) (1). At this time, photosynthetic bacteria were still considered to be plants and thus were also recognized to fix CO_2 for growth (2).

Nearly a century later, it was proposed that CO_2 fixation also plays a role in maintaining redox balance in the group of anoxygenic phototrophs known as purple nonsulfur bacteria (PNSB). PNSB obtain energy from light and carbon from organic substrates to support growth under anaerobic conditions (photoheterotrophic growth). In 1933, Muller reported that CO_2 had to be supplied for PNSB bacteria to grow photoheterotrophically on butyrate, a substrate more reduced than biomass (Table 1) (3). Muller proposed a hypothesis that CO_2 is needed to accept excess electrons and allow butyrate to be oxidized to the redox state of biomass (Fig. 1) (3). This hypothesis went untested for nearly 50 y. In 1977 Hillmer and Gest demonstrated that *Rhodobacter capsulatus* would grow on butyrate without CO_2 if growth conditions permitted N_2 fixation and therefore expulsion of electrons as H_2 (Fig. 1) (4). Richardson et al. (5) later showed that adding other alternative electron acceptors, such as dimethyl sulfoxide, allowed *R. capsulatus* to grow photoheterotrophically on butyrate without added CO_2 (Fig. 1). These studies confirmed that the role of CO_2 fixation during photoheterotrophic growth on butyrate is to accept electrons because other electron acceptors could substitute for CO_2 and permit growth.

In the early 1990s the groups of Kaplan and Tabita showed that CO_2 was required for photoheterotrophic growth even on substrates that are more oxidized than biomass (Table 1) (6, 7). *R. sphaeroides* mutants lacking ribulose 1,5-bisphosphate carboxylase (Rubisco), the key CO_2-fixing enzyme of the Calvin-Benson-Bassham cycle (Calvin cycle), were incapable of photoheterotrophic growth on succinate and malate. Photoheterotrophic growth was restored when the alternative electron acceptor dimethyl sulfoxide was supplied (6, 7). The fact that PNSB grow on oxidized organic substrates as the sole carbon source suggests that CO_2 derived from the oxidation of these substrates can support the necessary amount of electron-accepting CO_2 fixation. However, the reason for the requirement of CO_2 fixation for photoheterotrophic growth on substrates more oxidized than biomass and the quantitative contribution of CO_2 fixation to growth under these conditions were unclear.

Here we used ^{13}C-metabolic flux analysis (^{13}C-MFA) to quantify the significance of CO_2 fixation as an electron-accepting process during photoheterotrophic growth. The PNSB *Rhodopseudomonas palustris* was provided with acetate, a compound slightly more oxidized than biomass (Table 1), as its sole carbon source. Our results show cells oxidized 22% of the carbon in acetate to CO_2, and most of this then reentered metabolism via the Calvin cycle. This degree of Calvin cycle flux enabled cells to reoxidize almost half of the reduced cofactors generated during acetate oxidation. We also examined the effects of H_2 production by *R. palustris* on Calvin cycle flux and gene expression.

Results

Wild-Type Cells Have a Large Metabolic Flux from Acetate Through the Calvin Cycle. *R. palustris* wild-type strain CGA009 was grown anaerobically in light in mineral medium containing NH_4^+, a condition in which H_2 production via nitrogenase is repressed. To determine metabolic flux distributions, we first confirmed that cells convert acetate carbon and electrons entirely into biomass and CO_2 (Table 2). We then manually constructed a metabolic network from the *R. palustris* CGA009 genome sequence (8) (Table S1). We grew *R. palustris* with acetate labeled with ^{13}C in one of its carbons and measured mass isotopomer distributions in proteinaceous amino acids (Table S2). This information along with measurements of CO_2 efflux (Table 2) and biomass composition (Table S3) (both relative to acetate uptake)

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The authors declare no conflict of interest.

Data deposition: Microarray data have been deposited in the Gene Expression Omnibus (GEO) database, http://www.ncbi.nlm.nih.gov/geo (accession nos. GSM159274 and GSM159275).

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Table 1. Oxidation/reduction values of *R. palustris* growth substrates and biomass

Compound	Formula	Oxidation/reduction value[†]
Malate	$C_4H_6O_5$	+2
Succinate	$C_4H_6O_4$	+1
Acetate	$C_2H_4O_2$	0
Biomass[‡]	$CH_{1.8}N_{0.18}O_{0.38}$	−0.5
Butyrate	$C_4H_8O_2$	−2

[†]Values were determined as described by Gottschalk (39) and in *SI Methods*.
[‡]Based on the elemental composition of *R. palustris* 42OL (40).

were applied to the metabolic network using 13C-Flux software (9) to solve for intermediary metabolic fluxes. Details of the fitting solutions are described in the supplementary materials. The complete set of net and exchange flux values are listed in Table S4.

The flux map indicates that acetate is mainly assimilated via the glyoxylate shunt (Fig. 2*A*, red). The distribution of fluxes then seems to be primarily determined by the demand for biosynthetic precursors, with little flux through the lower TCA cycle (Fig. 2*A*, blue) or pyruvate dehydrogenase (Fig. 2*A*, *PDH*/*POR*). An exception to this trend is the flux through the Calvin cycle. Calvin cycle flux was quantified from the unique metabolite labeling patterns that it generated, such as fully labeled serine (Fig. S1). Flux through the CO_2-fixing enzyme of the Calvin cycle, Rubisco, was considerable at 29.6 mol% of the acetate uptake rate [0.58 μmol × mg dry cell weight $(DCW)^{-1} \times h^{-1}$; Fig. 2*A*]. The magnitude of the CO_2-fixing Rubisco flux was surprising because no CO_2 or $NaHCO_3$ was provided to the cultures. The sum of the flux through all CO_2-producing reactions was 43.8 mol% of the acetate uptake rate, indicating that approximately 22% of the acetate carbon was released as CO_2. Rubisco then fixed 67% of this CO_2 into cell material. This high Rubisco flux makes *R. palustris* metabolism efficient with respect to carbon utilization. According to the flux distribution, only 6% of acetate carbon is expelled as CO_2 (11.9 mol% of the acetate uptake rate), leaving 94% of the acetate carbon for incorporation into cell material (Fig. 2*A*).

Whereas most of the acetate carbon was used for biosynthesis, only approximately half (54%) of the reduced redox cofactors generated during acetate oxidation were oxidized in biosynthetic reactions (Table 3). Thus, to maintain redox balance and metabolic flow, *R. palustris* must recycle the remaining redox cofactors by some other means. The Calvin cycle can maintain cellular electron carriers in the oxidized reduction state after fixing CO_2 by oxidizing NADH via GAPDH. At 59.5 mol% of the acetate uptake rate (Fig. 2*A*), GAPDH flux was 44% of the total flux through reactions that generate reduced cofactor (Table 3). Thus, the Calvin cycle fulfilled a major role in maintaining redox balance by recycling reduced redox cofactors. Approximately 2% of reduced cofactors were oxidized by a proposed pyruvate–ferredoxin oxidoreductase flux (Fig. 2 and Table 3, *PDH*/*POR*).

Calvin Cycle Fluxes Are Diminished when Cells Produce H_2. H_2 production by nitrogenase, which produces H_2 along with NH_3 as an obligatory aspect of its catalytic cycle (10), enables photoheterotrophic growth of PNSB on highly reduced substrates (4). We used ^{13}C-MFA to investigate the effect of H_2 production on carbon fluxes from acetate and to determine how the Calvin cycle responds to an alternative means of electron disposal. For this purpose we decided to use an *R. palustris nifA** mutant, which expresses nitrogenase genes constitutively and thus produces H_2 via nitrogenase in NH_4^+-containing mineral medium, so that we could compare non–hydrogen-producing (wild-type) and hydrogen-producing (NifA*) cultures grown under the same conditions.

NifA is a σ^{54}-dependant enhancer-binding protein that is the master transcriptional regulator of nitrogenase genes (11). Previously described *nifA** mutants have single point mutations in the Q-linker region that is between the NifA N-terminal GAF domain and a central AAA^+ domain preceding the C-terminal helix-turn-helix DNA-binding domain (Fig. 3) (12). These mutations apparently enable the protein to adopt an active conformation in NH_4^+-grown cells. Because our original *nifA** point mutants can revert to the wild-type phenotype with a single nucleotide change, we established a stable *nifA** mutant by deleting 48 bp of the Q-linker, for use in our ^{13}C-MFA experiments. This NifA* strain, CGA676, produced H_2 at comparable levels to the previously described *nifA** point mutant, CGA584 (12).

Central metabolic flux distributions of the H_2-producing NifA* strain were remarkably similar to those in the non–H_2-producing wild type with the exception of fluxes through the Calvin cycle (Fig. 2). Although a similar amount of acetate (23%) was oxidized to CO_2 as in the wild type, the NifA* strain's Rubisco flux was

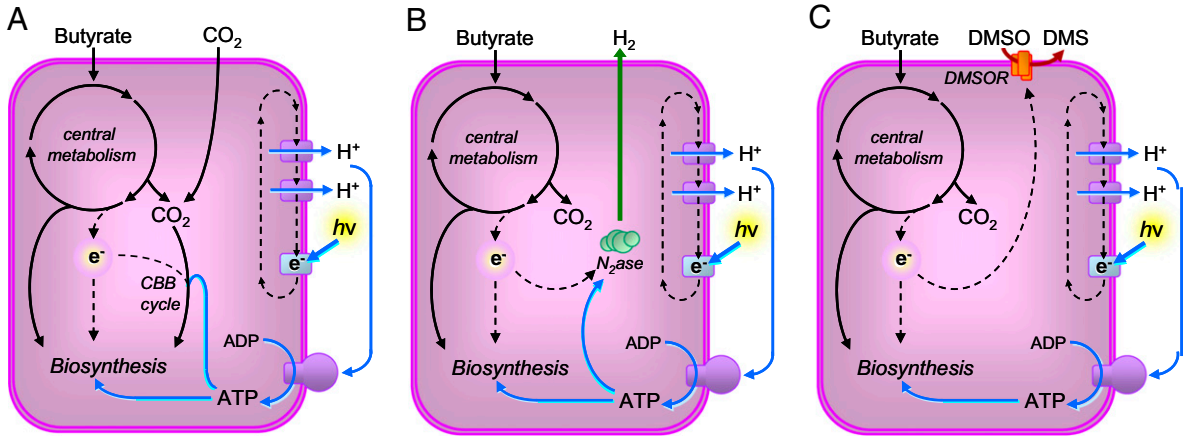

Fig. 1. Redox balancing strategies used by PNSB during anaerobic photoheterotrophic growth on a reduced carbon source. During photoheterotrophic growth of PNSB on butyrate, ATP is produced by cyclic photophosphorylation, in which electrons are repeatedly energized by light, allowing them to be cycled through a proton-pumping electron transport chain, rather than being transferred to a terminal electron acceptor. Thus, PNSB must use other means to dispose of extra electrons that are generated during use of a carbon source like butyrate, which is more reduced than biomass (Table 1). Three mechanisms are depicted: (*A*) CO_2 fixation via the Calvin cycle; (*B*) H_2 production via nitrogenase (N_2ase; green arrow); and (*C*) DMSO reduction by DMSO reductase (DMSOR; red arrow). Solid black lines, carbon metabolism; solid blue lines, energy metabolism; dotted lines, electron metabolism.

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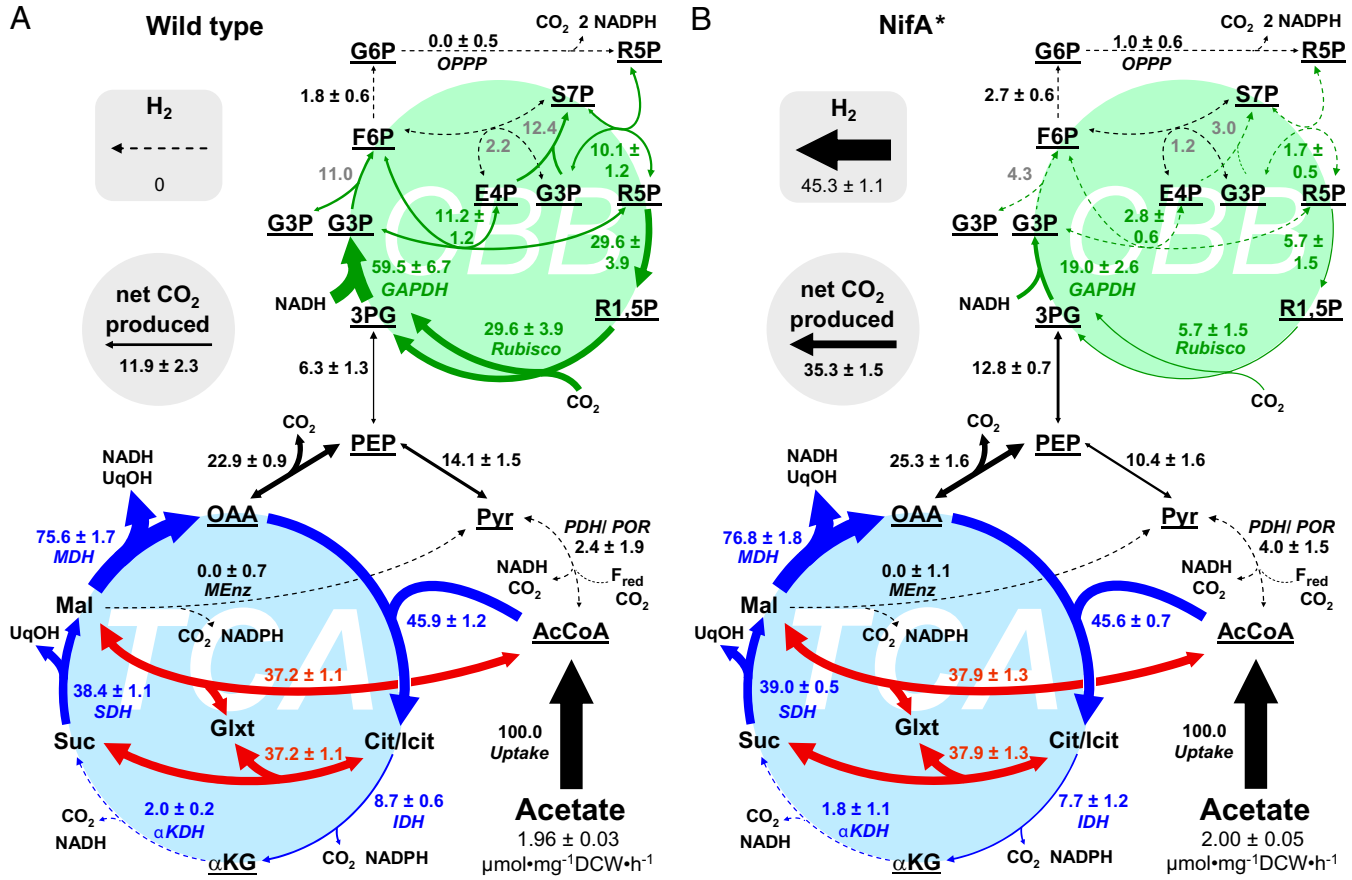

Fig. 2. Central metabolic fluxes in wild-type and NifA* *R. palustris* strains during photoheterotrophic growth on acetate. Both strains were grown with NH_4^+ as a nitrogen source. Under these conditions the wild type does not synthesize nitrogenase or produce H_2 (*A*), but the NifA* strain does synthesize nitrogenase and produce H_2 (*B*). Net fluxes values are listed as mole percentage of the acetate uptake rate ± 90% confidence intervals. Actual acetate uptake rates are below the uptake arrow. Gray values indicate that the confidence interval was larger than the mean. For brevity, biosynthetic fluxes and exchange fluxes (reverse fluxes) that could be determined are shown in Table S4. Net flux direction is indicated by an enlarged arrowhead. Flux magnitudes are indicated by arrow thickness. Fluxes that have a value of less than 5 are given a dashed arrow. The names of flux reactions not mentioned in the text are listed in Table S1.

approximately one fifth of that of the wild type (specific in vivo activity of 0.11 μmol × mg $DCW^{-1} \times h^{-1}$), and it thus refixed only 13% of CO_2. The low Rubisco flux during H_2 production resulted in a 3-fold increase in net CO_2 production compared with the wild type, leaving 82% of the carbon for biosynthesis (Fig. 2).

Similar to the wild type, nearly half of the reduced redox cofactors (47%) generated during acetate oxidation were used for biosynthesis during H_2 production by the NifA* strain (Table 3). Unlike the wild type, the Calvin cycle played a lesser role in maintaining redox balance during H_2 production. GAPDH flux was one third of the wild-type flux (Fig. 2), recycling only 14% of reduced redox cofactors (Table 3). H_2 production (0.45 ± 0.01 mol% of the acetate uptake rate) played a larger role in maintaining redox balance, recycling 34% reduced redox cofactors. Thus the NifA* strain used a combination of CO_2 fixation and H_2 production to recycle approximately half of its redox cofactor (one sixth and one third, respectively). As in the wild type, there was minor participation (3%) by the *PDH/POR* flux in maintaining redox balance. The difference between wild-type and NifA* GAPDH flux accounted for 89% of the electrons needed to produce the observed H_2 (Table 3). In other words, most of the electrons for H_2 production were diverted away from the Calvin cycle. The remaining electrons for H_2 production are accounted for by the NifA* strain's lower NADPH requirement to support its lower growth yield (Table 2). H_2 production had little effect on fluxes through other dehydrogenases or oxidoreductases (Table 3).

Table 2. Conversion of acetate to biomass, CO_2, and H_2 by *R. palustris*

		Yield (mol • mol acetate^{-1})				
Strain	g (h)	$Y_{Biomass}$[†]	Y_{CO_2}	Y_{H_2}	C recovery (%)	e$^-$ recovery (%)[‡]
Wild-type	8.4 ± 0.6	1.75 ± 0.15	0.11 ± 0.02	0	93 ± 8	98 ± 9
NifA*	9.4 ± 0.6	1.57 ± 0.08	0.34 ± 0.04	0.41 ± 0.05	96 ± 5	99 ± 5

Cultures were grown in minimal medium with 20 mM sodium acetate and NH_4^+ as the nitrogen source. Values are averages from 8 to 10 biological replicates ± SD.

[†]Moles of *R. palustris* biomass were determined from the elemental composition of *R. palustris* 42OL (40): $CH_{1.8}N_{0.18}O_{0.38}$, which has a molecular weight of 22.426 g/mol.

[‡]Electron recovery was based on available hydrogen as described by Gottschalk (39) and in *SI Methods*.

Table 3. Redox accounting based on fluxes through redox reactions

Reaction	Cofactor made	Wild type	NifA*
Biosynthesis NADH[†]	NADH	12.2	10.7
Biosynthesis NADP+[†]	$NADP^+$	−72.3	−63.2
αKDH	NADH	1.9	1.8
IDH	NADPH	8.7	7.7
SDH	QOH	38.4	39.0
MDH	NADH/QOH	75.6	76.8
MEnz	NADPH	0.0	0.0
PDH/POR	NAD^+/ Fd_{ox}	−2.4	−4.0
GAPDH (Calvin cycle)[‡]	NAD^+	−59.5	−19.0
OPPP	NADPH	0.0	1.0
H_2	Fd_{ox}	0.0	−45.3
Total cofactor reduction[§]		136.8	137.0
Total cofactor oxidation[§]		−134.2	−131.5
Redox balance (%)[¶]		98	96

Values are fluxes as mole percentages of acetate uptake rate. Italicized reactions are the same as those in Fig. 2 and in Table S1. Positive values indicate flux through reactions that form reduced redox cofactors (e.g., NADH), and negative values indicate flux through reactions that form oxidized redox cofactors (e.g., NAD^+). QOH, quinol; Fd_{ox}, oxidized ferredoxin.
[†]NADH and $NADP^+$ formation expected from biosynthesis based on the *R. palustris* biomass composition (Table S3).
[‡]NAD^+ is assumed to be made by GAPDH rather than $NADP^+$, on the basis of ref. 20.
[§]Total reduction: sum of fluxes through reactions forming reduced cofactors. Total oxidation: of fluxes through reactions forming oxidized cofactors.
[¶]Total oxidation ÷ total reduction × 100%.

Calvin Cycle Flux Is Essential for Growth in the Absence of H_2 Production. The NifA* strain maintained a low level of Calvin cycle flux, even though it disposed of most of its excess electrons by H_2 production. To assess the physiological importance of this low Calvin cycle flux, we deleted the genes encoding both type I and type II Rubiscos in the NifA* and wild-type strains. Consistent with previous work with *R. sphaeroides* (6, 7) and *R. palustris* (13), the Rubisco mutant of wild-type *R. palustris* (CGA669) was unable to grow photoheterotrophically in NH_4^+-containing medium (Table 4). However, the Rubisco mutant could grow photoheterotrophically in NH_4^+-containing medium when H_2 production was permitted by introducing the CGA676 *nifA** mutation to form strain CGA679. H_2 production seemed to compensate for the lack of Calvin cycle flux in expelling excess electrons, because the H_2 yield increased 1.3-fold in the NifA* Rubisco mutant strain compared with the NifA* strain with functional Rubisco enzymes (Table 4). However, the NifA* Rubisco strain had a 1.7-fold longer generation time compared with the NifA* strain (Table 4). The slower growth rate may be explained by the fact that the specific H_2 productivity was the same for both strains, indicating that nitrogenase activity did not increase despite the extra redox burden in the absence of Calvin cycle flux (Table 4). Thus, H_2 production can maintain redox balance in the absence of the Calvin cycle, but the growth rate is likely limited by the slow rate at which electrons are expelled as H_2 via nitrogenase.

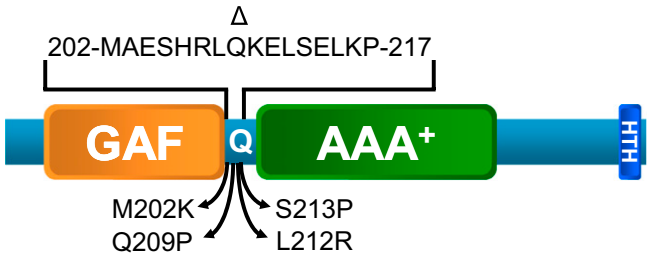

Fig. 3. Diagram of NifA structural domains and sites of NifA* mutations. A 16-aa NifA Q-linker region deletion strain (CGA676) produces H_2 in the presence of NH_4^+. This deletion mutant is much more stable than the NifA* point mutants that were previously described (12) and are shown below the NifA diagram. NifA and its domains are drawn to scale.

Additional Features of *R. palustris* Metabolism Revealed by 13C-MFA. There are some general features of the ^{13}C-metabolic flux maps outside of our main objectives that are worth mentioning briefly. First, *R. palustris* used a citramalate-dependent pathway (14) to synthesize approximately 94% of its isoleucine, whereas the rest was made by the traditional threonine-dependent pathway (Fig. S2). Second, flux between pyruvate and acetyl-CoA was reversible, suggesting that *R. palustris* has an active pyruvate ferredoxin oxidoreductase in addition to pyruvate dehydrogenase, which is generally considered to unidirectionally decarboxylate pyruvate to acetyl-CoA (Fig. 2 and Fig. S1). Third, there was little flux through NADPH-generating reactions (Table 3), suggesting that cells generated nearly 90% of their NADPH by oxidizing other redox cofactors (e.g., NADH oxidation via membrane-bound transhydrogenase encoded by RPA4180-2).

Electron Flux Distribution Is Regulated at the Level of Transcription. Our data show that electrons are diverted away from the CO_2-fixing Calvin cycle toward nitrogenase to produce H_2. However, the flux maps do not indicate whether nitrogenase simply outcompetes the Calvin cycle for electrons or whether regulatory mechanisms are involved in the redistribution of electrons. We used quantitative PCR (qPCR) to compare expression levels of *nifD*, encoding the nitrogenase α-subunit, and several Calvin cycle genes in the wild-type and NifA* strains. As expected, *nifD* expression levels were much higher (450-fold) in the NifA* strain compared with the wild type because NH_4^+ does not repress nitrogenase expression in the NifA* strain (Fig. 4). Similar to the Calvin cycle flux trend, Calvin cycle gene expression levels were lower in the H_2-producing NifA* strain than in the non–H_2-producing wild type. Thus, the shift of electrons from CO_2 fixation to H_2 production is controlled at the transcriptional level. We also compared the transcript levels of genes encoding central metabolic enzymes in five different *nifA** mutants vs. the wild-type strain, all grown on acetate with NH_4^+ (12). Like the flux maps, the only consistent changes in transcripts encoding central metabolic enzymes in response to H_2 production were for those encoding Calvin cycle enzymes, which were down-regulated (Table S5). A similar decrease in Calvin cycle transcript levels was also observed in wild-type cultures grown under conditions permitting N_2 fixation, and therefore H_2 production, relative to non–H_2-producing cultures (Table S5).

RegSR Does Not Transcriptionally Control the Calvin Cycle in Response to H_2 Production. In other PNSB, the redox-sensing two-component regulatory system RegBA/PrrBA modulates gene expression of both Calvin cycle operons (reviewed in refs. 15 and 16). We therefore examined the effect of deleting the *R. palustris regBA* homologs, *regSR*, on Calvin cycle gene expression. For these experiments we used wild-type *R. palustris* CGA010 (a derivative of CGA009) and its preexisting Δ*regSR* mutant, CGA2023. We used the presence and absence of NH_4^+ to respectively repress or activate H_2 production by nitrogenase in these strains. When not producing H_2, the Δ*regSR* mutant had 3.1-fold higher *cbbL* expression than the wild type and negligible increases in *cbbM* and *cbbP* expression (Fig. 5). The Δ*regSR* mutation had no effect on the expression of the three Calvin cycle genes or on *nifD* expression during H_2 production (Fig. 5). Most importantly, *regSR* was not required for the drastic difference in Calvin cycle *cbbL*, *cbbM*, and *cbbP* expression observed between H_2-producing cultures and non–H_2-producing cultures (Fig. 5), indicating that this two-component system is not central to integrating redox metabolism in *R. palustris* as it is in other PNSB (15, 16).

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Table 4. Effect of Rubisco mutations on growth and H_2 production

Strain	Genotype	g (h)	H_2 yield (mol × mol ace^{-1})	Sp. H_2 productivity (mmol × g DCW^{-1} × h^{-1})†	H_2:biomass (mol × g protein^{-1})
CGA009	Wild type	7.4 ± 0.2	None detected	–	–
CGA669	Δ*cbbLS*::Km^R Δ*cbbM*	No growth	–	–	–
CGA676	*nifA**	8.3 ± 0.8	0.43 ± 0.04	1.12 ± 0.08	66 ± 5
CGA679	*nifA** Δ*cbbLS*::Km^R Δ*cbbM*	12.8 ± 0.8‡	0.58 ± 0.03‡	1.02 ± 0.07	92 ± 3‡
CGA679 complemented	*nifA** Δ*cbbLS*::Km^R Δ*cbbM* (pBBP*cbbLSX*)	9.8 ± 1.4	0.44 ± 0.08	0.98 ± 0.04	68 ± 11

Cultures were grown in minimal medium with 20 mM sodium acetate and NH_4^+ as the nitrogen source. Values are averages ± SDs from 3 to 5 biological replicates.

†Specific H_2 productivity. Determined using the Monod model as described in ref. 21.

‡Significantly different from CGA676 characteristic (Student's *t* test, two-tailed, equal variance, $P < 0.05$).

Discussion

CO_2 Fixation Is a Major Aspect of Photoheterotrophic Metabolism. Our results illustrate why, and to what extent, CO_2 fixation is required during photoheterotrophic growth on substrates, like acetate, that are more oxidized than *R. palustris* biomass. During the oxidation of acetate, approximately twice as much reduced redox cofactor is produced than is needed for biosynthesis. CO_2 fixation plays a major and obligatory role as an electron-accepting process to recycle the majority of these excess reduced cofactors under non–N_2-fixing conditions. The NifA* strain, which synthesizes nitrogenase constitutively, recycled approximately half of its electrons using a combination of CO_2 fixation and H_2 production. Because no other electron acceptors were available, our results suggest that *R. palustris* growing on acetate can only devote approximately half of its reduced cofactor toward biosynthesis, and the rest must be recycled by other means (e.g., CO_2 fixation and H_2 production). These ^{13}C-labeling experiments highlight two underappreciated facts: (*i*) that photoheterotrophic growth is associated with a challenge in maintaining redox balance, and (*ii*) that the CO_2-fixing Calvin cycle is not only at the heart of photoautotrophic metabolism but can be a central aspect of photoheterotrophic metabolism as well. Our results are in agreement with theoretical work predicting that photoheterotrophic growth is associated with a large Rubisco flux (17).

Interplay Between CO_2 Fixation and H_2 Production as Redox-Balancing Mechanisms. Because the NifA* strain is capable of both CO_2 fixation and H_2 production, we were able to examine the interplay between these two redox-balancing processes. In the NifA* strain electrons were shifted away from CO_2 fixation toward H_2 production (Table 3). Although Calvin cycle flux decreased significantly in response to H_2 production, it did not shut off completely. When Calvin cycle flux was prevented in the NifA* Rubisco strain, CGA679, the growth rate was severely impaired (Table 4). Interestingly, the specific H_2 production rate did not vary when Rubisco activity was eliminated (Table 4), suggesting that there is a bottleneck preventing an increase in the H_2 production rate. Possibilities are a rate limitation in electron transfer to nitrogenase, nitrogenase concentration, or the rate of nitrogenase activity. It seems likely that the NifA* Rubisco strain's growth rate is limited by the rate at which redox balance can be achieved by expelling electrons as H_2. When the Calvin cycle is also available, the *nifA** mutant may be able to adjust electron distribution between the two redox-balancing mechanisms to maximize its growth rate.

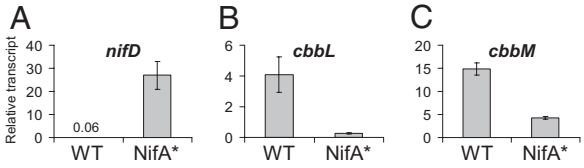

Fig. 4. Effect of H_2 production on Calvin cycle gene expression. Relative levels of transcripts between wild-type *R. palustris* that is not producing H_2 (CGA009; WT) and a mutant (CGA676; NifA*) that produces H_2 in the presence of NH_4^+. The two strains were grown under identical conditions with 20 mM sodium acetate with NH_4^+ present. (*A*) *nifD*, encoding the α-subunit of nitrogenase; (*B*) *cbbL*, encoding the Rubisco type I large subunit; and (*C*) *cbbM*, encoding Rubisco type II. Data are averages from duplicate cultures, with error bars representing the range. Transcript levels are shown relative to *fixJ* transcript levels.

We observed that the shift from CO_2 fixation to H_2 production was accompanied by a decrease in Calvin cycle gene transcription (Figs. 4 and 5). It was previously reported that in *R. sphaeroides*, Rubisco transcript levels varied in response to other factors affecting the intracellular redox state, such as feeding substrates of differing redox state (18) and adding the electron acceptor dimethyl sulfoxide (7). Although we have excluded the RegSR system as being central to controlling Calvin cycle fluxes in *R. palustris* in response to redox changes due to H_2 production, there are a number of other regulatory mechanisms that could be involved. For example, high NADH levels can stimulate Calvin cycle phosphoribulokinase activity (19, 20), which in turn can stimulate Calvin cycle gene expression through the transcriptional regulator CbbR (15).

Alternative Roles of the Calvin Cycle and Nitrogenase in Nature. Use of CO_2 fixation for redox balance is not restricted to photoheterotrophic metabolism. For example, succinate-producing fermentative bacteria fix CO_2 via C_3-carboxylating enzymes as part of a redox-balancing mechanism (21). It is possible that there are niches, for example, microaerobic environments, where it is advantageous for nonphotosynthetic, obligately heterotrophic bacteria to expend some ATP and use the Calvin cycle to maintain redox balance. Complete Calvin cycle operons have been annotated in heterotrophic bacteria for which autotrophic growth has not been demonstrated, including *Dechloromonas aromatica* str. RCB (22), *Methylibium petroleiphilum* PM1 (23), and *Burkholderia xenovorans* (24). Rubisco genes have also been detected in environmental isolates related to *Arthrobacter*, *Bacillus*, and *Streptomyces* (25). Calvin cycle activity by such bacteria may help to explain observations in which adding acetate to soil in the dark can stimulate CO_2 fixation (26).

As proposed at least as early as 1962 by Ormerod and Gest (27), H_2 production by PNSB may also have an underappreciated role in maintaining redox balance. In PNSB mutants having constitutive nitrogenase activity, nitrogenase expression decreased in response to the addition of the electron acceptor dimethyl sulfoxide or the expression of Calvin cycle genes from a plasmid (28, 29). These results suggest that there are mechanisms in place to control the contribution of nitrogenase toward redox balance in response to other electron acceptors. However, the use of nitrogenase for

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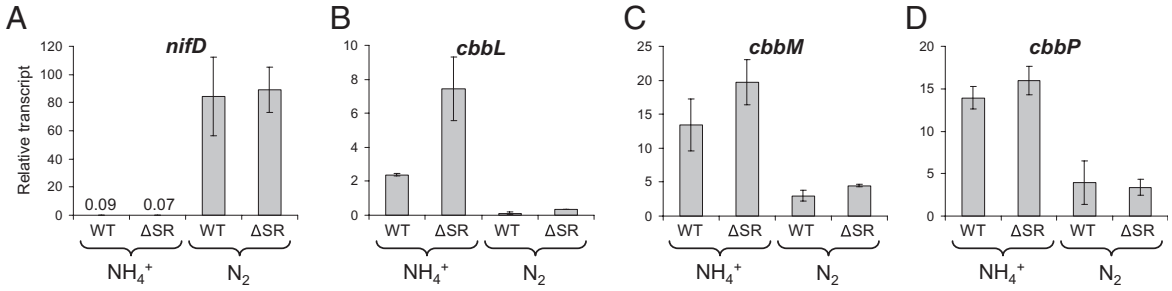

Fig. 5. Calvin cycle gene expression in a Δ*regSR* strain. Relative levels of transcripts between wild-type *R. palustris* (CGA010; WT) and a Δ*regSR* mutant derivative (CGA2023; ΔSR). H_2 production was accomplished by supplying N_2 as the sole nitrogen source. Acetate was the carbon source. Data are averages from duplicate cultures, with error bars representing the range. (*A*) *nifD*, encoding the α-subunit of nitrogenase; (*B*) *cbbL*, encoding the Rubisco type I large subunit; (*C*) *cbbM*, encoding Rubisco type II; and (*D*) *cbbP*, encoding phosphoribulokinase. Transcript levels are shown relative to *fixJ* transcript levels.

redox balance is likely an auxiliary or specialized mechanism because the large ATP requirement of nitrogenase would present challenges for its use in maintaining redox balance in a nonphotosynthetic heterotroph.

Methods

Chemicals, Bacteria, and Culture Conditions. Sodium [1-^{13}C] and [2-^{13}C]acetate were purchased from Cambridge Isotope Laboratories. All experiments were conducted with *R. palustris* wild-type strains CGA009 or CGA010 and their derivatives (Table S6). CGA009 is defective in uptake hydrogenase activity, whereas the uptake hydrogenase mutation is repaired in CGA010 (30). CGA010 uptake hydrogenase was not functional in the studies herein owing to insufficient nickel in the growth medium. *R. palustris* was grown anaerobically in front of a 60-W light bulb at 30 °C in 16.5-mL volumes of defined mineral medium (31) in sealed 27-mL anaerobic culture tubes (Bellco) without added $NaHCO_3$ or CO_2. Cultures were grown in minimal medium either with $(NH_4)_2SO_4$ under an Ar headspace (i.e., PM) or without $(NH_4)_2SO_4$ under a N_2 headspace (i.e., NFM). Sodium acetate or disodium succinate was supplied at a final concentration of 20 mM or 10 mM, respectively. For comparisons between the *nifA** parent, CGA676, and its Rubisco mutant derivatives, cultures were grown aerobically in the dark in PM, before being adapted and tested in anaerobic PM in the light. *Escherichia coli* strains DH5α and S17-1 were grown at 37 °C in Luria-Bertani medium. Ampicillin, kanamycin, and gentamcyin were supplied at 100, 100, and 20 μg/mL, respectively, for *E. coli*, and kanamycin and gentamycin were each supplied at 100 μg/mL for *R. palustris* where appropriate.

Strain Construction. In-frame deletion mutants of *R. palustris* were generated using primers and plasmids listed in Table S6 as previously described (30), with the modifications described in *SI Methods*. To complement the *cbbLS* deletion, *cbbLSX* were expressed from a plasmid downstream of the promoter region for RPA0944, which encodes GAPDH, as described in *SI Methods*.

Analytical Techniques. Acetate was quantified as previously described (32) using a Varian HPLC equipped with a UV detector. H_2 was quantified using gas chromatography as previously described (12). Total CO_2 [i.e., the sum of headspace CO_2 (g) plus dissolved CO_2 (aq)] was quantified according to previously described methods (33), with details in *SI Methods*.

***R. palustris* Biomass Composition.** Measurements were made for strains CGA009 and CGA676 in PM as previously described (21), with details in *SI Methods*.

^{13}C-Labeling Experiments. CGA009 and CGA676 were grown in PM containing [1-^{13}C] or [2-^{13}C]acetate, such that there were three biological replicates of each strain for each substrate isotopomer. Cultures were transferred once in midexponential phase to avoid light-limiting conditions at high cell densities and to dilute the unlabeled inoculum to below 1% of the harvested biomass. Upon reaching a density between 0.3 and 0.4 OD_{660}, cultures were chilled in ice water and centrifuged at 4 °C. Cell pellets were lysed by sonication, soluble protein hydrolyzed, and amino acids purified, quantified, and derivatized with *tert*-butyldimethylsilyl groups, as previously described (21). Derivatized amino acids were analyzed by gas chromatography–mass spectrometry using an Agilent 5973 GC-MSD equipped with a DB5-MS column (30 m × 0.25 mm × 0.25 μm) as previously described (21).

Metabolic Flux Analysis. Metabolic fluxes were determined using 13C-Flux software (9). Flux models were fit to experimental data using the 13C-Flux program, Donlp2 (P. Spellucci, Technische Universität Darmstadt, Germany). Mass isotopomer distributions were corrected for the natural abundances of amino acid atoms other than carbon and all atoms in the derivatization groups using previously described software (34). Datasets resulting from experiments using either [1-^{13}C] or [2-^{13}C]acetate were fit against a single metabolic model as previously described (35). To determine whether multiple flux solutions existed, ≈100 different starting values were used for Donlp2 runs. Confidence intervals for individual fluxes were determined as previously described (36). The isotopomer measurements used to determine specific fluxes are given in Table S7, with examples illustrated in Fig. S1.

Microarray and qPCR. Microarray analyses were carried out with an Affymetrix *R. palustris* Custom GeneChip as previously described (12) for CGA009 grown in NFM with 20 mM acetate and were deposited at http://www.ncbi.nlm.nih.gov/geo under accession nos. GSM159274 and GSM159275. For qPCR experiments, cultures were grown to a density of 0.3–0.5 OD_{660} with unlabeled acetate. RNA for microarrays and qPCR was isolated as previously described (37). cDNA was generated as previously described (12). The qPCR reactions were set up as previously described (38), except *R. palustris* chromosomal DNA was used to generate standard curves, and transcript levels were normalized to transcript levels of *fixJ*. We also made use of previously reported microarray data (12).

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